



Chapter 7

Risk Analysis

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7 Risk Analysis

(5 hours)

7.1 Origin of risk in projects

7.2 Risk analysis of projects (Sensitivity, breakeven and scenario analyses)

7.3 Decision tree

Definition

- **Risk analysis** is the systematic process of identifying, measuring, and evaluating uncertainty in project cash flows to assess its impact on economic decision-making.
- It involves quantifying uncertainties (such as cost overruns, delays, or technical failures) to evaluate the financial viability of projects. It uses tools like sensitivity analysis, probability distributions, and Monte Carlo simulations to calculate the expected loss or return, helping stakeholders make informed decisions under uncertainty.
- Risk analysis is the process of identifying and evaluating uncertainties that could cause a project's actual financial return to differ from its expected return. It bridges the gap between deterministic estimates (fixed numbers) and the reality of fluctuating markets, technical hurdles, and shifting regulations.

Definition

Key Methods of Evaluating Risk

Engineers use several structured techniques to quantify and visualize how uncertainty impacts project viability:

- a. *Sensitivity Analysis*: Identifies which specific variables (e.g., fuel costs, labor rates) have the largest effect on the project's Net Present Value (NPV). Often visualized using Spider Diagrams or Tornado Diagrams.
- b. *Scenario Analysis*: Evaluates the project under set conditions, typically comparing a "Base Case" against "Best Case" (optimistic) and "Worst Case" (pessimistic) scenarios.
- c. *Break-Even Analysis*: Determines the critical value of a single variable (like sales volume or price) at which the project's NPV equals zero, indicating the point where it transitions from a loss to a profit.

Definition

- d. *Expected Value Analysis*: Assigns probabilities to different outcomes to calculate a weighted average return.
- e. *Monte Carlo Simulation*: A computer-based technique that runs thousands of "what-if" iterations using random samples from probability distributions to provide a range of possible outcomes and their likelihoods.
- f. *Decision Tree Analysis*: A graphical flowchart used for sequential decision-making, mapping out various choices and their probabilistic outcomes over time.

Origin of Risk in Projects

The origin of risk is fundamentally the lack of precise knowledge regarding future conditions. It stems from the inherent uncertainty in project variables such as cash flows, market demands, and technological performance.

These origins are typically categorized into two primary spheres:

I. Internal Origins (Controllable)

- These arise from within the project team or the host organization. Management has a higher degree of influence over these factors.
- Estimation Inaccuracy: Errors in forecasting project life, initial capital investment, or operating costs.
- Management & Human Factors: Poor communication, unrealistic deadlines, inexperienced personnel, or inadequate resource allocation.
- Technical & Design: Use of unproven technology, design flaws, or system incompatibilities.
- Operational Inefficiency: Process delays, equipment failures, or labor disputes within the organization.

Origin of Risk in Projects

2. External Origins (Uncontrollable)

These stem from the broader environment outside the organization's direct control and often require adaptation or contingency planning rather than prevention.

- **Market & Economic Shifts:** Sudden changes in consumer demand, competitor actions, interest rates, inflation, or currency fluctuations.
- **Legal & Regulatory:** New environmental standards, safety laws, or changes in government policy that affect project feasibility.
- **Environmental & Force Majeure:** Natural disasters (floods, earthquakes) or geopolitical instability (strikes, civil unrest).
- **Stakeholder Dynamics:** Changes in requirements from the client, suppliers failing to deliver, or lack of public acceptance.

Origin of Risk in Projects

Origin of Risk	Description	Typical Causes	Impact on Project
Technical Risk	Risk due to engineering and technology uncertainties	Design errors, unproven technology, equipment failure	Rework, cost overrun, performance failure
Economic Risk	Risk arising from changes in economic conditions	Inflation, price escalation, interest rate change	Increased project cost, reduced profitability
Financial Risk	Risk related to financing and cash flow	Delay in funds, high loan interest, poor financial planning	Project delay, liquidity problems
Market Risk	Risk due to uncertainty in demand and price	Demand fluctuation, competition, price variation	Lower revenue, reduced ROI
Managerial Risk	Risk caused by poor management and organization	Inefficient planning, weak coordination, lack of skills	Schedule delay, inefficiency

Origin of Risk in Projects

Natural Risk	Risk due to natural disasters	Earthquake, flood, landslide, cyclone	Damage to structures, time overrun
Political Risk	Risk arising from political instability	Policy change, tax change, government change	Project suspension, cost increase
Legal Risk	Risk due to legal and contractual issues	Contract disputes, land acquisition issues	Litigation, delay
Social Risk	Risk from public and community issues	Public opposition, resettlement problems, strikes	Work stoppage, delay

Risk analysis of projects (Sensitivity, breakeven and scenario analyses)

Break-even analysis is a financial decision-making tool used to determine the exact point where total revenue equals total costs. At this Break-Even Point (BEP), a project or product generates zero profit and incurs zero loss. In engineering contexts, it is fundamental for evaluating the viability of new designs, equipment purchases, and "make-buy" decisions.

Cost Components

Fixed Cost (FC): Cost independent of production volume (e.g., initial investment, land, rent, insurance)

Variable Cost (VC): Cost proportional to level of output (e.g., material, labor per unit)

Selling Price per Unit: The price at which each unit is sold to customers.

Break- Even Analysis

In terms of output (units):

$$\text{BEP} = \frac{FC}{SP - VC}$$

where,

SP = Selling price per unit

VC = Variable cost per unit

FC = Fixed Cost

Break- Even Analysis

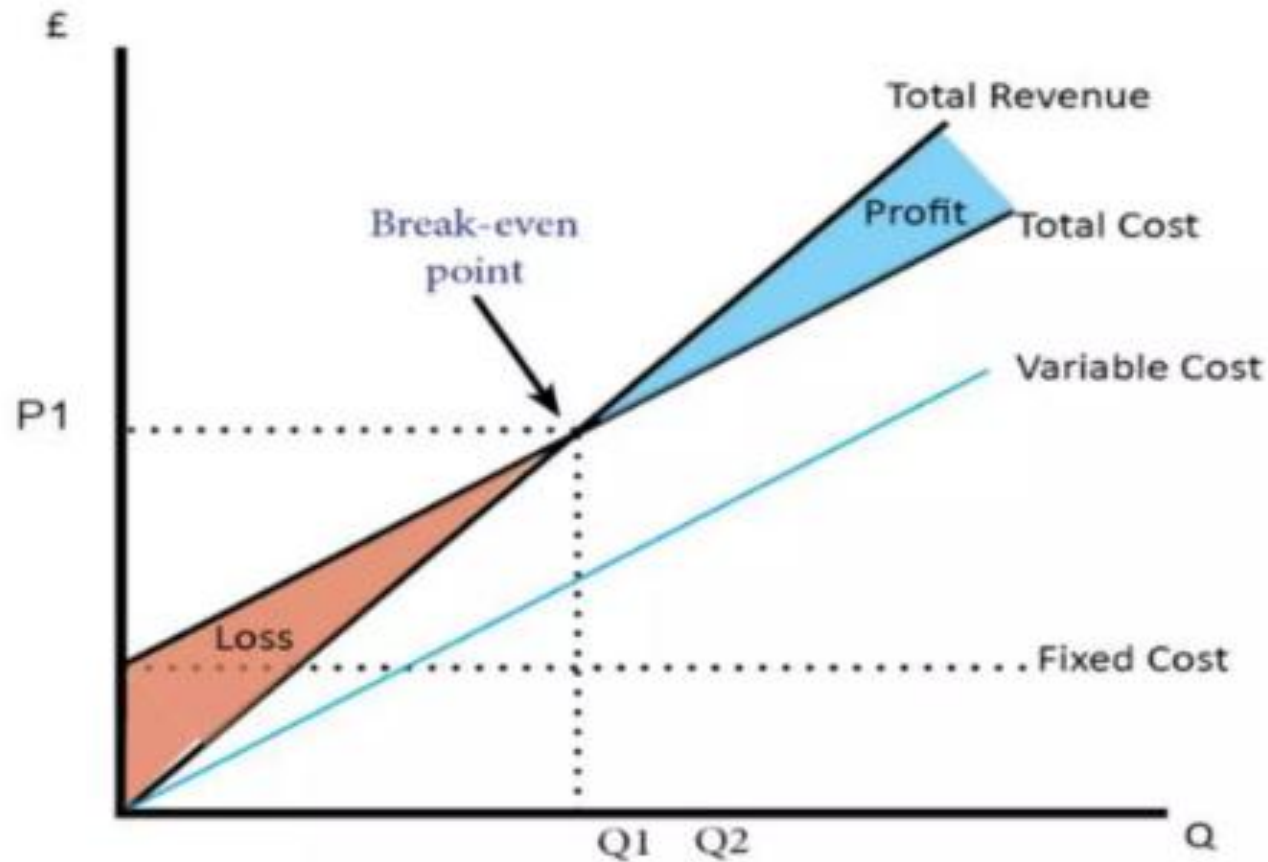


Fig : Visual Representation / BEP chart

X-axis: Production volume (units)

Y-axis: Cost and revenue

Fixed cost line: Horizontal

Total cost line: Starts from fixed cost with slope equal to variable cost

Revenue line: Starts from origin with slope equal to selling price

Intersection point represents the break-even point

Break- Even Analysis

- Left of BEP: Project operates at a loss
- At BEP: Project is economically just viable
- Right of BEP: Project earns profit
- Steeper revenue line: Higher selling price
- Higher fixed cost: Break-even shifts right (more risk)

Break- Even Analysis

Total Cost

$$C_T = C_F + C_V$$

Total cost = Fixed cost + Variable cost

Total Variable Cost

$$C_V = V_c \times Q$$

Variable cost per unit $\times$ Quantity

Total Revenue
(Sales)

$$S_T = S_p \times Q$$

Selling price $\times$ Quantity

Contribution per
Unit

$$C_u = S_p - V_c$$

Profit contribution from each unit

Contribution Ratio
(P/V ratio)

$$\frac{S_p - V_c}{S_p}$$

Contribution as % of selling price

Break- Even Analysis

Break-Even
Condition

$$S_T = C_T$$

Revenue equals total cost

Break-Even Output
(Units)

$$Q_{BEP} = \frac{C_F}{S_p - V_c}$$

Units required for zero profit

Break-Even Sales
(Revenue)

$$S_{BEP} = \frac{C_F}{\text{Contribution Ratio}}$$

Sales value at break-even

Output for Desired
Profit

$$Q = \frac{C_F + P}{S_p - V_c}$$

Units required for target profit

Profit Formula

$$\text{Profit} = (S_p - V_c)Q - C_F$$

General profit equation

Break- Even Analysis

Break-Even Operating Hours	$X = \frac{A_2 - A_1}{B_1 - B_2}$	Break-even hours between two alternatives
Capacity Utilization at BEP	$\frac{Q_{BEP}}{Q_{max}} \times 100\%$	% capacity used at BEP
Margin of Safety (Units)	$\text{Actual Sales} - Q_{BEP}$	Safe zone above BEP
Margin of Safety (%)	$\frac{\text{Actual Sales} - Q_{BEP}}{\text{Actual Sales}} \times 100$	Risk indicator

Break- Even Analysis

- Calculate break–even hours of operation per year to become cost equal and recommended economic pump if it is to be operated 5 hours daily at full load.

Particulars	KHASA Pump	SARVO Pump
Capacity	100 hp	100 hp
Purchase cost (Rs.)	5,00,000	10,00,000
Tax per year (Rs.)	10,000	15,000
Maintenance cost per year (Rs.)	36,500	29,200
Efficiency	80%	90%
Life year	5	5
Salvage value	20% of purchase cost	20% of purchase cost
MARR	20% per year	20% per year
Electricity cost	Rs. 10/kWh	Rs. 10/kWh

Break- Even Analysis

KHASA Pump:

$$\begin{aligned} \text{CR} &= 5,00,000 (\text{A/P}, 20\%, 5) - 1,00,000 (\text{A/F}, 20\%, 5) \\ &= 5,00,000 (0.3344) - 1,00,000 (0.1344) \\ &= 1,67,200 - 13,440 \\ &= 1,53,760 \end{aligned}$$

$$\text{Maintenance cost/year} = 36,500$$

$$\text{Tax per year} = 10,000$$

$$\text{Operating cost} = (100 \times 0.746 \times 10 \times X) / 0.8 = 932.5X$$

$$\text{Total cost equation} = 2,00,260 + 932.5X \dots(i)$$

Break- Even Analysis

SARVO Pump:

$$\begin{aligned} CR &= 10,00,000 (A/P, 20\%, 5) - 2,00,000 (A/F, 20\%, 5) \\ &= 3,07,520 \end{aligned}$$

$$\text{Maintenance cost/year} = 29,200$$

$$\text{Tax per year} = 15,000$$

$$\text{Operating cost} = 828.88X$$

$$\text{Total cost equation} = 3,51,720 + 828.88X \dots(ii)$$

Equating equation (i) and (ii)

$$2,00,260 + 932.5X = 3,51,720 + 828.88X$$

Break- Even Analysis

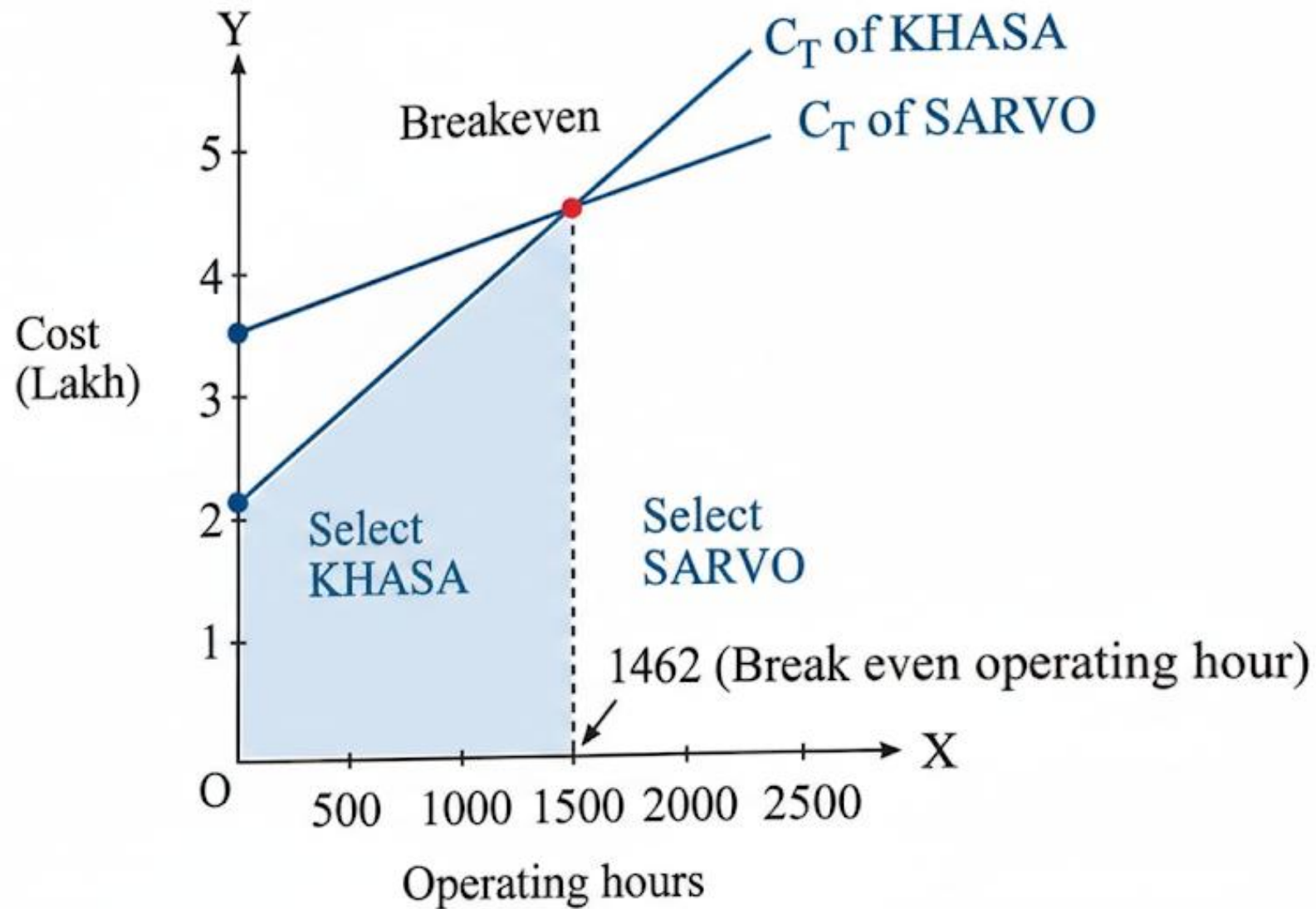
or,

$$103.62X = 151460$$

$$\therefore X = 1,461.687 \approx 1,462 \text{ hours}$$

$$\text{Yearly operating hours} = 5 \times 365 = 1,825 \text{ hours.}$$

Break- Even Analysis



Break- Even Analysis

➤ Calculate breakeven volume of a cable manufacturing company from the following data:

Total Cost = Rs. 12,00,000

Variable Cost = Rs. 4,00,000

Income from sale = Rs. 15,00,000 at production of 5,000 units , plot the BEP chart.

Given,

Total cost (C_t) = 12,00,000

Total variable cost (C_v) = 4,00,000

Total sales (S_t) = 15,00,000

We know output (Q) = 5,000 units

Break- Even Analysis

We know,

$$C_t = C_F + C_v$$

Or,

$$C_F = C_t - C_v$$

Or,

$$C_F = 12,00,000 - 4,00,000$$

$$\therefore C_F = 8,00,000$$

Similarly, Variable cost per unit (v_c) = C_v / Q

$$v_c = 4,00,000 / 5,000 = 80 / \text{unit}$$

Break- Even Analysis

$$\text{Selling price per unit } (S_p) = S_t / Q = 15,00,000 / 5,000 = 300 / \text{unit}$$

$$\text{Breakeven output volume} = C_F / (S_p - V_c)$$

$$\therefore \text{Breakeven output volume} = 8,00,000 / (300 - 80) = 3,636.36 \text{ units}$$

At 3,636.36 output level;

$$S_t = 3,636.36 \times 300 = 10,90,909$$

$$C_t = 3,636.36 \times 80 + 8,00,000 = 10,90,909$$

$$\therefore S_t = C_t$$

Break- Even Analysis

- A machine has a Fixed Cost of Rs. 40,00,000. It has variable cost Rs. 45,000 per unit. Find BEP both in volume and value if selling price per unit is Rs. 60,000. What would be the effect on profit/loss when fixed cost increase by 10% and selling price decreases by 5%.

Solution:

$$\text{BEP} = \frac{C_F}{S_p - V_c} = \frac{40,00,000}{60,000 - 45,000}$$

= 266.6667 units

$$C_T = C_F + C_V = C_F + V_c(X)$$

$$= 40,00,000 + 45,000 \times 266.66667$$

$$= 40,00,000 + 1,20,00,000$$

$$= 1,60,00,000$$

$$S_T = S_p(X) = 60,000 \times 266.66667 = 1,60,00,000$$

$$\therefore C_T = S_T = 1,60,00,000$$

Break- Even Analysis

If C_F by 10%;

$$C_F = 40,00,000 \times 1.1 = 44,00,000$$

$$C_T = C_F + C_V = 44,00,000 + 1,20,00,000 = 1,64,00,000$$

If $S_p \downarrow$ by 5%; $S_p = 60,000 \times 0.95 = 57,000$

$$\Rightarrow S_t = S_p \times (X)$$

$$= 57,000 \times 266.66667 = 1,52,00,000$$

$$\text{Profit or loss} = S_t - C_t = 1,52,00,000 - 1,64,00,000 = -12,00,000$$

$$\therefore \text{Loss} = 12,00,000$$

Break- Even Analysis

➤ The details of production costs and revenues of a project are as under:

Total cost = 85,000

Fixed cost = 40,000

Sales (Rs.) = 1,05,000

Sales volume = 15,000

Find breakeven point in terms of number of units. What should be the output if the profit desired is Rs. 50,000?

The breakeven point in terms of number of units

We know,

Total cost (C_t) = Fixed cost (C_F) + Variable cost (C_v)

$$C_v = C_t - C_F$$

$$= 85,000 - 40,000$$

$$C_v = \text{Rs. } 45,000$$

Break- Even Analysis

Variable cost per unit (V_c) = $45,000 / 15,000 = \text{Rs. } 3 \text{ per unit}$

Selling price per unit (S_p) = $1,05,000 / 15,000 = \text{Rs. } 7 \text{ per unit}$

So,

$Q_{BEP} = C_F / (S_p - V_c) \text{ units}$

$= 40,000 / (7 - 3) \text{ units}$

$= 10,000 \text{ units}$

Break- Even Analysis

Similarly, if the profit desired is Rs. 50,000

Profit = Total sales – Total cost

$$50,000 = S_p \times Q - (C_F + C_v)$$

$$50,000 = S_p \times Q - (40,000 + V_c \times Q)$$

$$50,000 = S_p \times Q - 40,000 - V_c \times Q$$

$$90,000 = (S_p - V_c) Q$$

$$90,000 = (7 - 3) Q$$

$$Q = 22,500$$

Sensitivity Analysis

Sensitivity analysis is a technique used to determine how the output of a decision, model, or project changes when one or more input variables are varied.

sensitivity analysis is a powerful tool used to evaluate how changes in economic assumptions influence project profitability and feasibility. It ensures better and safer engineering investment decisions under uncertainty.

In real-life engineering and economic problems:

- Costs may increase
- Selling price may decrease
- Demand may fluctuate
- Interest rate may change

Since these inputs are uncertain, sensitivity analysis helps us:

- Measure risk
- Identify critical variables
- Make better decisions under uncertainty

Sensitivity Analysis

Purpose

- To identify the most critical variable
- To measure risk in economic decisions
- To test the robustness of a project
- To compare alternatives under uncertainty

Sensitivity Analysis

➤ Draw sensitivity chart using PW formulation of the following cash information. It is desired to evaluate the sensitivity of PW of $\pm 30\%$ changes on:

i) Interest rate

ii) Investment

When; $I = 10000$

AR = 4,000 for the first year and increases by 500 for next 4 year

MARR = 10%

Setting the given information in PW formation:

$$PW = -10,000 + 4,000 (P/A, 10\%, 5) + 500 (P/G, 10\%, 5)$$

$$PW = -10,000 + 4,000 (P/A, 10\%, 5) + 500 (P/G, 10\%, 5)$$

$$PW = -10,000 + 4,000 \times 3.7908 + 500 \times 6.8618$$

$$PW = -10,000 + 15,163 + 3,431 = 8,595$$

Sensitivity Analysis

Interest

Let change in interest rate by $\pm 30\%$

If $i = +30\%$;

$$\begin{aligned}PW &= -10,000 + 4,000 (P/A, 13\%, 5) + 500 (P/G, 13\%, 5) \\&= -10,000 + 4,000 \times 3.5172 + 500 \times 6.1802 \\&= 7,159\end{aligned}$$

If $i = -30\%$;

$$\begin{aligned}PW &= -10,000 + 4,000 (P/A, 7\%, 5) + 500 (P/G, 7\%, 5) \\&= -10,000 + 4,000 \times 4.1002 + 500 \times 7.76947 \\&= 10,298\end{aligned}$$

Sensitivity Analysis

i) Investment

Let change in initial investment (I) by $\pm 30\%$

If I = +30%;

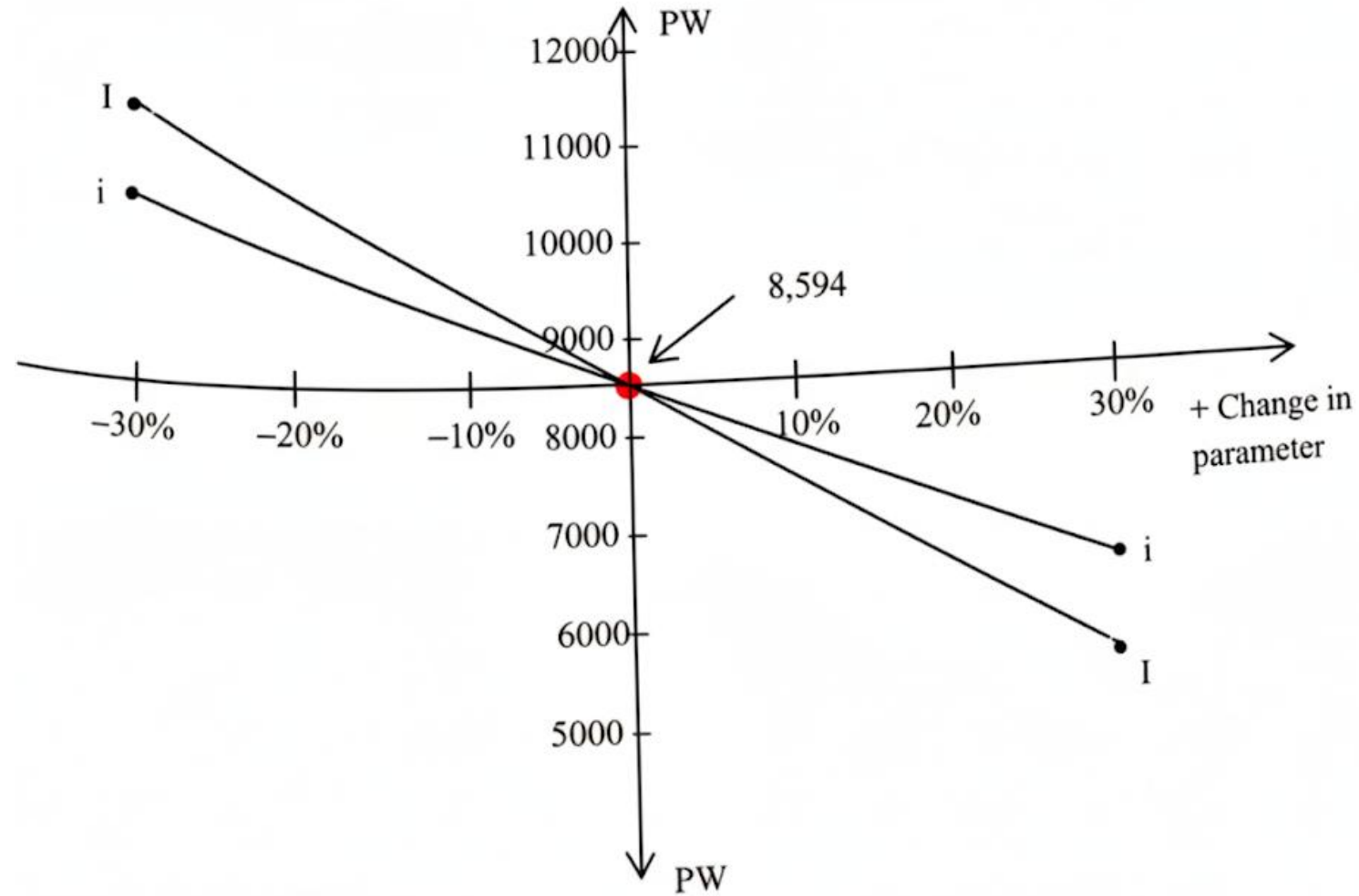
$$\begin{aligned}PW &= -10,000 (1.3) + 15,163 + 3,431 \\&= -13,000 + 18,594 \\&= 5,594\end{aligned}$$

If I = -30%;

$$\begin{aligned}PW &= -10,000 (0.7) + 15,163 + 3,431 \\&= -7,000 + 18,594 \\&= 11,594\end{aligned}$$

Parameters	+30%	0%	-30%
Interest (i)	7,159	8,595	10,298
Investment (I)	5,594	8,594	11,594

Sensitivity Analysis



Scenario Analysis

Scenario analysis is a decision-making technique used to evaluate how a project performs under different combinations of assumptions.

Unlike sensitivity analysis (which changes one variable at a time), scenario analysis changes multiple variables simultaneously to represent realistic future conditions

Instead of asking:

“What happens if interest rate changes?”

We ask:

“What happens if interest rate increases, operating cost rises, and demand falls together?”

Each combination is called a scenario.

Scenario Analysis

Types of Scenarios

1. Base Case Scenario

- Most likely values of all variables
- Normal expected condition

2. Best Case Scenario

- High revenue
- Low cost
- Favorable interest rate
- High demand

3. Worst Case Scenario

- Low revenue
- High cost
- High interest rate
- Low demand

Scenario Analysis

Sensitivity Analysis	Scenario Analysis
One variable at a time	Multiple variables together
To find which variable affects the result most	To see how the project performs under different situations
Change only interest rate by $\pm 10\%$	Change interest rate, cost, and sales together
Less realistic	More realistic
Simple	Slightly complex
Shows impact of single factor	Shows overall good/bad conditions

Scenario Analysis

- From the following information, calculate NPV for each scenario by assuming $I = 1,25,000$, $MARR = 15\%$ & life of project is 5 years.

Variable considered	Worse-case scenario	Most-likely case	Best-case scenario
Unit demand /yr	1,600	2,000	2,400
Unit price (Rs)	48	50	53
Variable cost (Rs)/unit	17	15	12
Fixed cost (Rs)/yr	11,000	10,000	8,000
Salvage value (Rs)	30,000	40,000	50,000

Scenario Analysis

(a) Worse case scenario:

$$\begin{aligned}\text{NPV (15\%)} &= -1,25,000 + (1,600 \times 48 - 1,600 \times 17 - 11,000) (P/A, 15\%, 5) + \\ &30,000 (P/F, 15\%, 5) \\ &= -1,25,000 + 38,600 (3.3522) + 30,000 (0.4972) \\ &= -1,25,000 + 1,29,395 + 14,916 \\ &= 19,311 \text{ (Profit)}\end{aligned}$$

(b) Most likely case scenario:

$$\begin{aligned}\text{NPV (15\%)} &= -1,25,000 + (2,000 \times 50 - 2,000 \times 15 - 10,000) (P/A, 15\%, 5) + \\ &40,000 (P/F, 15\%, 5) \\ &= -1,25,000 + 60,000 (3.3522) + 40,000 (0.4972) \\ &= -1,25,000 + 2,01,132 + 19,888 \\ &= 96,020 \text{ (Profit)}\end{aligned}$$

Scenario Analysis

(c) Best case scenario:

$$\text{NPV (15\%)} = -1,25,000 + (2,400 \times 53 - 2,400 \times 12 - 8,000) (P/A, 15\%, 5) + 50,000$$

(P/F, 15%, 5)

$$\text{NPV (15\%)} = 2,02,899 \text{ (Profit)}$$

Scenario analysis indicates, there is no risk for investment.

Decision Tree Analysis

The **Decision Tree** method in risk analysis is a visual and quantitative tool used to map out various decision paths, their uncertain outcomes, and the resulting financial or utility impacts. It is a staple technique in project management, particularly within the Perform Quantitative Risk Analysis process.

Core Components of a Decision Tree

A standard tree is drawn from left to right and consists of several key symbols:

- **Decision Nodes (Squares):** Represent points where a choice must be made between two or more alternatives.
- **Chance Nodes (Circles):** Represent uncertain events or "risks" that may occur after a decision.
- **Branches:** Lines connecting nodes that represent different paths or outcomes.
- **End/Terminal Nodes (Triangles):** Mark the final outcome of a specific path, often showing a payoff like profit or loss.

Decision Tree Analysis

The Analytical Process

1. Structuring the Tree: Identify the core decision and all subsequent alternatives and chance events.
2. Assigning Probabilities: For every branch from a chance node, assign a probability (e.g., 60% chance of success). These must sum to 100% for each node.
3. Determining Payoffs: Assign a monetary value or cost to each terminal node.
 - Calculating Expected Monetary Value (EMV): Use a "folding back" technique from right to left:
 - For chance nodes, multiply each outcome by its probability and sum them.
 - For decision nodes, select the path with the best EMV.
 - Formula:

$$EMV = \sum (Probability \times Payoff)$$

Decision Tree Analysis

Symbol	Meaning
□ Square	Decision node (choice to be made)
○ Circle	Chance node (uncertain event)
→ Branch	Possible outcome
Payoff	Profit or loss at end

Purpose in Engineering Economics

- Market demand is uncertain
- Project success/failure is possible
- Technology performance is uncertain
- Investment decisions involve risk

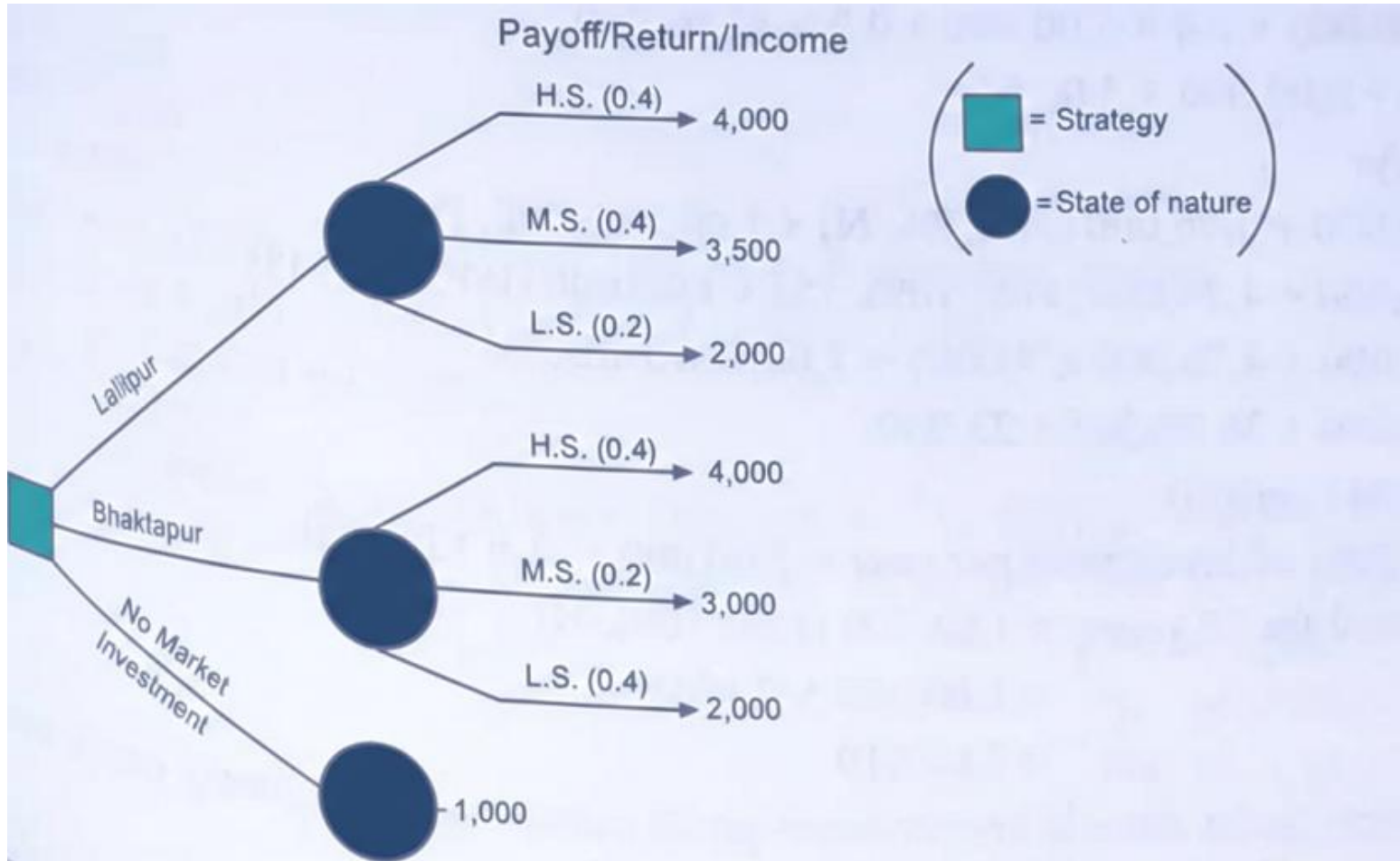
Decision Tree Analysis

- A business firm is considering expanding its branch in either Bhaktapur or Lalitpur. From a preliminary survey, the following data has been obtained:

Business nature	Lalitpur Probability	Income (000)	Bhaktapur Probability	Income (000)
Low success	0.2	2,000	0.4	2,000
Medium success	0.4	3,500	0.2	3,000
High success	0.4	4,000	0.4	4,000

The cost of advertising for Lalitpur is Rs. 1,50,000 and for Bhaktapur is Rs. 2,00,000. Find which branch the firm should prefer.

Decision Tree Analysis



Decision tree diagram

Decision Tree Analysis

From the decision tree diagram:

Expected Return/Income from Lalitpur

$$= 0.4 \times 4,000 + 0.4 \times 3,500 + 0.2 \times 2,000$$

$$= 34,00,000$$

$$\text{Expected net profit from Lalitpur} = 34,00,000 - 1,50,000 = 32,50,000$$

Decision Tree Analysis

- *Expected Return/Income from Bhaktapur*

$$= 0.4 \times 4,000 + 0.2 \times 3,000 + 0.4 \times 2,000$$

$$= 30,00,000$$

$$\text{Expected net profit from Bhaktapur} = 30,00,000 - 2,00,000 = 28,00,000$$

Since expected net profit from Lalitpur is greater than that of Bhaktapur, the firm should prefer to expand branch in Lalitpur.

Assignment

1) Perform sensitivity analysis of the following project over a range of $\pm 30\%$ at an interval of $\pm 10\%$ in: (i) Initial investment

(ii) Net annual revenue

(iii) Useful life ,Use the Present Worth (PW) formulation.

Initial investment (Rs.) = 1,00,000

Net annual revenue (Rs.) = 40,000, Salvage value (Rs.) = 15,000

Useful life (years) = 6 , MARR (%) = 10

2) Perform sensitivity analysis by IRR using FW formulation (with increment of 10%) over a range of $\pm 30\%$ in:

(i) Useful life

(ii) MARR

Take MARR = 10%.

Project	First cost	Annual benefit	Annual expenses	Salvage value	Useful life
A	Rs. 3,00,000	Rs. 1,50,000	Rs. 25,000	10% of first cost	10 years

Assignment

- 3) A company produces an electronics timing switch used in consumer and commercial products manufactured by other firms. The fixed cost and total cost are Rs. 40,000 and Rs. 85,000, respectively. The total sales are Rs. 1,05,000, and the sales volume is 15,000 units. i) Determine the breakeven point in terms of number of units. ii) What should be the output (units) if the desired profit is Rs. 50,000?
- 4) If the cost of 25 watt CFL bulb is Rs. 260 whereas the cost of 100 watt filament bulb is Rs. 35 but these bulbs have equal lighting power, which bulb do you prefer in your use and why? When electricity cost is Rs. 11 per unit (kWh-hr).

Assignment

5) Perform sensitivity analysis of the following project over a range of $\pm 30\%$ in:

- i) Initial investment
- ii) Net annual revenue
- iii) Useful life year

Draw also sensitivity diagram.

Initial investment = 5,00,000

Net annual revenue = 1,20,000

Salvage value = 80,000

Useful life year = 6

MARR = 10%

Assignment

6) A newly established hat company estimated the following data:

Output of hats per annum = 3,00,000 units

Expected sales revenue per annum = Rs. 1,50,00,000

Fixed costs = Rs. 35,00,000

Variable cost = Rs. 66,00,000

Find:

i. Break-even output level.

ii. If fixed cost increases to Rs. 40,00,000; its effect on break-even output.

iii. What should be the output if net profit is desired Rs. 10,00,000 per year?

7) Suppose fixed cost associated with the production of a product is Rs. 6,00,000. Assume variable cost is Rs. 35,000 and the selling price is Rs. 30,000 for each product and maximum sales amount is Rs. 40,00,000 when sales volume is 100 units. Determine the break-even point for this situation and plot the BEP chart.

Thank you for your attention!